

Maintaining your System: - Automation

We're going to talk about cron process, but more specifically, how we interact with via use of crontabs. Crontabs is one of processes that is started during boot, which is responsible for facilitating and managing cron jobs.

Crontab is special file with formatting that is recognised by cron process to execute each line step by step. Crontab requires 6 specific values:-

Value	Description
MIN	what minute to execute at
Hour	what hour to execute at
DOM	what day of month to execute at
MON	What month of year to execute at
DOW	what day of week to execute at
CMD	The Actual command that will be executed.

```
root@ip-10-49-115-207:~# crontab -l
Edit this file to introduce tasks to be run by cron.

Each task to run has to be defined through a single line
indicating with different fields when the task will be run
and what command to run for the task.

To define the time you can provide concrete values for
minute (m), hour (h), day of month (dom), month (mon),
and day of week (dow) or use '*' in these fields (for 'any').

Notice that tasks will be started based on the cron's system
daemon's notion of time and timezones.

Output of the crontab jobs (including errors) is sent through
email to the user the crontab file belongs to (unless redirected).

For example, you can run a backup of all your user accounts
at 5 a.m every week with:
0 5 * * * tar -czf /var/backups/home.tar.gz /home/

For more information see the manual pages of crontab(5) and cron(8)

# h dom mon dow   command
```

```
Wildcard when you don't care about that field.
Example lines and plain-English meanings:
1. * * * * * /usr/bin/echo "hello" -- Runs
every minute (every minute of every
hour/day/month/year) and prints "hello".
2. @ 0 * * * * /usr/bin/backup.sh -- Runs
daily at 00:00 (midnight) and runs the backup
script.
3. 30 3 * * * /usr/bin/weekly-report --
Runs every Monday at 03:30 AM to create a
weekly report.
4. @ 0 1 * * * /usr/bin/monthly-maintenance --
Runs at midnight on the 1st of every month
for monthly maintenance.
5. @ * 12 * * * cp -R /
/home/elastic/elastic /usr/backups/ --
Your example runs at minute 0 every 12 hours
(midnight, noon) to copy the Documents
folder to /usr/backups/.
6. @reboot /usr/bin/start-app -- Special
shortcut that runs the command once when
the system boots (equivalent to scheduling at
startup).
```

Why crontabs, it automates repetitive tasks so you don't have to run them manually (backups, maintenance, reports, monitoring). Important tip always use absolute paths in CMD, test commands manually first; and use crontab -e to edit current users crontab.

Quiz

1) When will the crontab on deployed instance (10.49.159.91) run?

A) @reboot

crontab -l command

```
Cron Job Generated (you may copy & paste it to your crontab)
# * * * * * cp -R /home/elastic/Documents/elasticsearch-repository-2-4-1

Your cron job will be run at: (5 times displayed)
- 2021-04-26 00:00:00 UTC
- 2021-04-26 12:00:00 UTC
- 2021-04-27 00:00:00 UTC
- 2021-04-27 12:00:00 UTC
- 2021-04-28 00:00:00 UTC

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Output of the crontab jobs (including errors) is sent through
email to the user the crontab file belongs to (unless redirected).

For example, you can run a backup of all your user accounts
at 5 a.m every week with:
0 5 * * * tar -czf /var/backups/home.tar.gz /home/

For more information see the manual pages of crontab(5) and cron(8)

# h dom mon dow   command
```

```
root@ip-10-49-159-91:~# crontab -l
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minute (m), hour (h), day of month (dom), month (mon),
and day of week (dow) or use '*' in these fields (for 'any').

Notice that tasks will be started based on the cron's system
daemon's notion of time and timezones.
```

```
# Output of the cronjob (including errors) is sent through
# email to the user the cronjob file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 2 a.m. every week with:
# 0 2 * * 1 tar -zcf /var/backups/home.tar /home/
#
# For more information see the manual pages of cron(8) and crontab(1)
#
# h dom mon dow   day   command
# mmm mm dd dd hh:mm:ss shell/command
*/ * * * * /usr/sbin/atd
```

Task 7:- Maintaining your System: Package Management

Introducing packages and Software Repos:-

When developers wish to submit software to the community, they will submit it to an "apt" repository. If approved, their programs and tools will be released into the wild. Two of the most redeeming features of Linux shine to light here: User accessibility and the merit of open source tools.

When using the `is` command on a Ubuntu 20.04 Linux machine, these files serve as the gateway/registry.

```
ubuntu@ip-10-10-29-121:~$ apt-get install is
apt.conf.d auth.conf.d preferences.d sources.list sources.list.d trusted.gpg.d
ubuntu@ip-10-10-29-121:~$ ls -la /etc/apt/sources.list.d/
total 12
-rw-r--r-- 1 root root 1024 Oct 10 10:10 sources.list
```

The following two lines to add software from Canonical's "partner" repository.

```
# This software is not part of Ubuntu, but is offered by Canonical and the
# respective vendors as a service to Ubuntu users.
deb http://archive.canonical.com/ubuntu bionic partner
deb-src http://archive.canonical.com/ubuntu bionic partner
```

Additional repositories can be added by using the `add-apt-repository` command or by listing another provider. For example, some vendors will have a repository that is closer to their geographical location.

Managing Your Repositories (Adding and Removing)

Normally we use the `apt` command to install software onto our Ubuntu system. The `apt` command is a part of the package management software also named `apt`. `apt` contains a whole suite of tools that allows us to manage the packages and sources of our software, and to install or remove software at the same time.

One method of adding repositories is to use the `add-apt-repository` command we illustrated above, but we're going to walk through adding and removing a repository manually.

Whilst you can install software through the use of package installers such as `dpkg`, the benefits of `apt` means that whenever we update our system — the repository that contains the pieces of software that we add also gets checked for updates.

In this example, we're going to add the text editor Sublime Text to our Ubuntu machine as a repository as it is not a part of the default Ubuntu repositories. When adding software, the integrity of what we download is guaranteed by the use of what is called GPG (GNU Privacy Guard) keys. These keys are essentially a safety check from the developers saying, "here's our software". If the keys do not match up to what your system trusts and what the developers used, then the software will not be downloaded.

So, to start, we need to add the GPG key for the developers of Sublime Text 3. (Note that TryHackMe instances do not have internet access and so we're not expecting you to add this to the machine that you deploy, as it would fail.)

1. Let's download the GPG key and use `apt-key` to trust it: `apt-key adv --keyserver keyserver.ubuntu.com --recv-keys 3FA9E26364730CE4A04F1C45BEC800000180AE61`
2. Now that we have added this key to our trusted list, we can now add Sublime Text 3's repository to our `apt` sources list. A good practice is to have a separate file for every different community/3rd party repository that we add.
3. Let's create a file named `sublime-text.list` in `/etc/apt/sources.list.d` and enter the repository information like so:


```
root@linux3:~# touch /etc/apt/sources.list.d/sublime-text.list
root@linux3:~# echo 'deb https://download.sublimetext.com/apt/stable/ bionic main' > /etc/apt/sources.list.d/sublime-text.list
```
- 3.1. And now use `dpkg` or a text editor of your choice to add & save the Sublime Text 3 repository into this newly created file.
- 3.2. After we have added this entry, we need to update `apt` to recognize this new entry — this is done using the `apt update` command.
- 3.3. Once successfully updated, we can now proceed to install the software that we have trusted and added to `apt` using `apt install sublime-text`.
- 3.4. Once successfully updated, we can now proceed to install the software that we have trusted and added to `apt` using `apt install sublime-text`.

Removing packages is as easy as reversing. This process is done by using the `add-apt-repository --remove` command or by manually deleting the file that we previously added to. Once removed, we can just use `apt remove [software-name-here]` (e.g. `apt remove sublime-text`).

Task 8:- Maintaining your System: Logs

We briefly touched upon log files and where they can be found in Linux fundamentals

part 1. Let's quickly recap - located in `/var/log` directory, these files and folders contain login information for applications and services running on your system.

The OS has become pretty good at automatically managing this logs in process that is known as "rotating".

I have highlighted some logs from 3 services running on Ubuntu machine:

* An Apache2 web server

* logs for fail2ban service, which is used to monitor attempted brute force example

* UFW service which is used as firewall.

```
ubuntu@ip-172-31-23-158:~$ ls -la /var/log
total 120
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log.1
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log.2.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log.3.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log.4.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log.5.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log.6.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 alternatives.log.7.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 amazon
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apache2
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.1
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.2.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.3.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.12.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.24.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.26.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.27.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.28.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.176.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.177.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.178.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.192.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.193.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.194.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.195.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.196.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.197.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.198.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.199.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.200.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.201.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.202.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.203.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.204.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.205.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.206.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.207.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.208.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.209.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.210.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.211.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.212.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.213.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.214.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.215.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.216.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.217.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.218.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.219.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.220.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.221.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.222.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.223.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.224.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.225.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.226.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.227.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.228.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.229.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.230.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.231.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.232.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.233.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.234.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.235.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.236.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.237.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.240.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.241.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.244.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.248.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.249.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.250.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.251.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.252.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.253.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.254.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.255.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.256.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.257.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.258.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.259.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.260.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.261.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.262.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.263.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.264.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.265.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.266.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.267.gz
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-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.269.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.270.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.271.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.272.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.273.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.274.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.275.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.276.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.277.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.278.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.279.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.280.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.281.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.282.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.283.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.284.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.285.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.286.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.287.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.288.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.289.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.290.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.291.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.292.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.293.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.294.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.295.gz
-rw-r--r-- 1 root root 12288 Oct 10 10:10 apt.log.296.gz
-rw-r--r
```

activity. Example two types of log files below are interest of :-

1) access log 2) error log

```
ubuntu@ip-172-31-23-158:/var/log/apache2$ ls
access.log      access.log.3.gz  error.log.1      error.log.4.gz
access.log.1    access.log.4.gz  error.log.10.gz  error.log.5.gz
access.log.10.gz access.log.5.gz  error.log.11.gz  error.log.6.gz
access.log.11.gz access.log.6.gz  error.log.12.gz  error.log.7.gz
access.log.12.gz access.log.7.gz  error.log.13.gz  error.log.8.gz
access.log.13.gz access.log.8.gz  error.log.14.gz  error.log.9.gz
access.log.14.gz access.log.9.gz  error.log.2.gz   other_vhosts_access.log
access.log.2.gz error.log         error.log.3.gz
```

Quiz! -

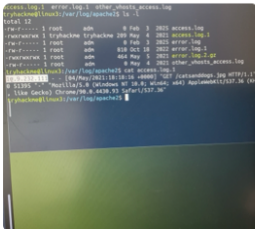
1) look for apache 2 logs on deployable linux machine.

A)

```
tryhackme@linux3: /var/log/apache2
File Edit View Search Terminal Help
tryhackme@linux3:/$ cd /var/log/apache2
tryhackme@linux3:/var/log/apache2$ ls
access.log      error.log         error.log.2.gz
access.log.1    error.log.1      other_vhosts_access.log
tryhackme@linux3:/var/log/apache2$
```

2) what is IP address of user who visited the site?

A)



3) what file did they access?

A)

```
tryhackme@linux3:/var/log/apache2$ cat access.log.1
10.9.232.111 - - [04/May/2021:18:18:16 +0000] "GET /catsanddogs.jpg HTTP/1.1"
0 51395 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML
, like Gecko) Chrome/90.0.4430.93 Safari/537.36"
tryhackme@linux3:/var/log/apache2$
```

catsanddogs.jpg

Continue your learning in **Bash Scripting and Regular Expressions**